

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: Artificial Intelligence
NAME: R. RUPA SREE

COURSE CODE: CSA1714
REGISTER NUMBER:192472122

COURSE OUTCOME COVERED

CO5: *Design AI-based systems using PySpark, RDD operations, and intelligent agent architectures, using scalable systems and integrate components in distributed AI applications.*
(BTL 5)

Assessment Tool Weightage: 20%

QUESTIONS: 5
Marks:50

Duration : 1 Hour
Total

Annexure A

Reverse Engineering

Code of Conduct:

I R. Rupa Sree (192472122) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

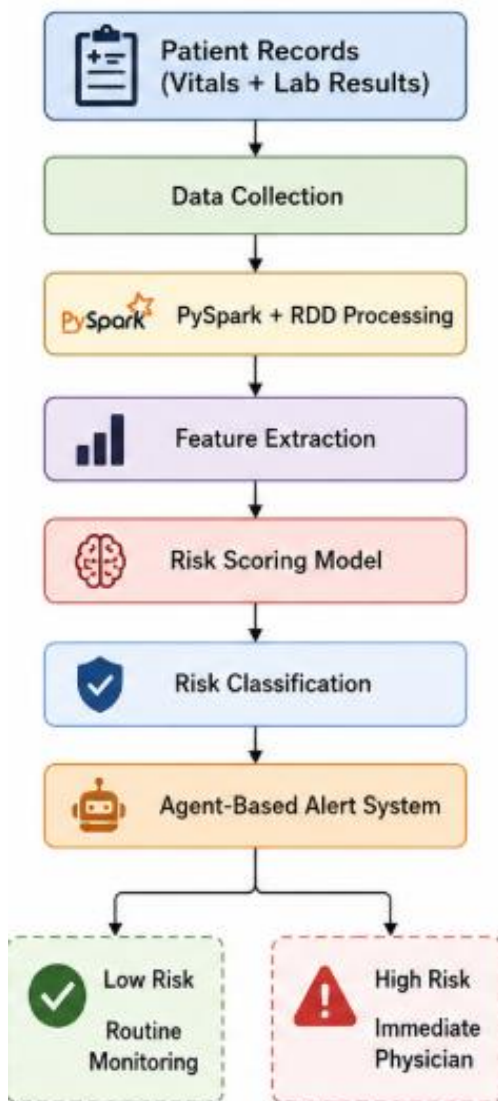
Signature of the student:
R. RUPA SREE

Reverse Engineering a Healthcare Diagnosis Assistant

A hospital's AI assistant analyses patient vitals and lab results to flag high-risk cases for immediate physician review. By observing its outputs and behavior, reconstruct the distributed AI pipeline using PySpark RDDs. Explain how feature extraction from patient records, risk scoring, and agent-based alert mechanisms are likely implemented.

ANS:

1.Distributed AI Pipeline



2. Feature Extraction

Patient records are processed using PySpark RDDs.

Important features include:

- Heart rate
- Blood pressure
- Temperature
- Oxygen level
- Blood glucose
- Hemoglobin
- Other abnormal lab values

Example:

Patient Record

↓

RDD.map()

↓

Extract Vital/Lab Features

↓

Feature Vector

3. Risk Scoring

The extracted features are given to a risk prediction model.

For example:

$$RiskScore = f(V, italsLabResults)$$

The score can be classified as:

Risk Score	Status
0–0.30	Low
0.31–0.70	Medium
0.71–1.00	High

A high score indicates that the patient requires immediate attention.

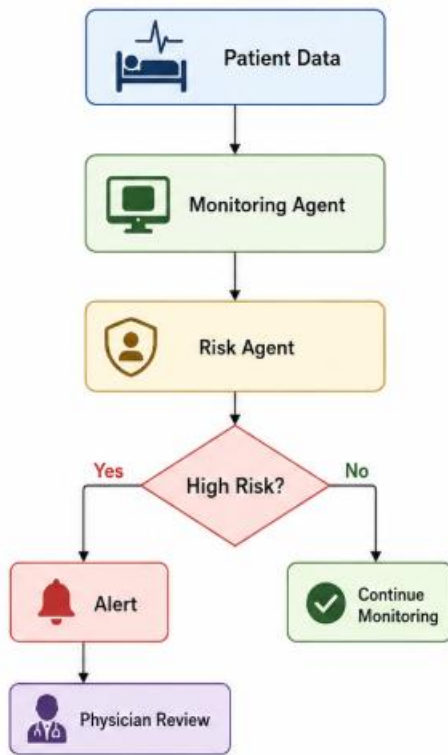
4. Agent-Based Alert Mechanism

Different intelligent agents can perform different tasks:

- Monitoring Agent: continuously monitors patient data.
- Risk Agent: calculates the risk score.
- Alert Agent: generates alerts for high-risk patients.
- Priority Agent: ranks critical cases for physician review.

5. PySpark RDD Operations

RDD Operation	Purpose
map()	Extract patient features
filter()	Identify abnormal records
flatMap()	Process multiple observations
reduceByKey()	Aggregate patient statistics
groupByKey()	Group records by patient



Conclusion

The healthcare assistant likely uses PySpark RDDs for distributed patient-data processing, feature extraction to identify important clinical indicators, and a risk-scoring model to detect high-risk cases. Intelligent agents then monitor results, prioritize critical patients, and automatically send alerts to physicians. This architecture provides scalable, real-time, and automated clinical decision support.

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Understanding of Industry Problem	20%	Comprehensive understanding of real-world problem and constraints	Good understanding with minor gaps in requirements	Basic understanding; some requirements unclear	Poor understanding; misinterprets problem context
Application of Engineering Concepts	25%	Applies advanced and relevant concepts effectively to solve the problem	Applies appropriate concepts with minor errors	Limited application of concepts	Incorrect or no application of concepts
Solution Design	25%	Innovative, practical, and feasible solution aligned with industry standards	Logical and feasible solution with minor gaps	Basic solution with limited feasibility	Unclear or impractical solution
Use of Tools	15%	Effective use of modern tools, technologies, and real data	Appropriate use of tools with correct implementation	Limited or inconsistent use of tools	Minimal or incorrect use of tools
Analysis	10%	Thorough analysis with validated results and performance evaluation	Good analysis with reasonable validation	Basic analysis with limited validation	Poor or no analysis
Professionalism	5%	Highly professional, well-documented, and clearly presented work	Good documentation and presentation	Basic documentation with some gaps	Poor documentation and presentation